

CineMR: Tool-Integrated Vision-Language Reasoning for Quantitative Cardiac MRI Assessment

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Abstract

Cardiovascular magnetic resonance (CMR), including cine imaging, is a reference standard for the noninvasive assessment of cardiac morphology and ventricular function. Cine CMR interpretation integrates qualitative visual assessment with quantitative measurements of ventricular volumes, ejection fraction, myocardial mass, wall thickness, and regional wall motion. Current medical vision-language models (VLMs) cannot reliably derive quantitative measurements from multidimensional cine images without analysis tools. We present **CineMR**, a tool-augmented VLM that invokes cardiac image-analysis tools and integrates their outputs into interleaved reasoning for quantitative CMR assessment. We also construct a multi-cohort visual question answering benchmark covering quantitative metric extraction, multiclass diagnosis, and differential diagnosis, together with tools for segmentation, phase selection, volumetry, morphometry, and regional wall motion analysis. CineMR is trained with supervised fine-tuning (SFT) on tool-interaction traces followed by Group Relative Policy Optimization (GRPO) with conditional tool-use rewards. On the multi-cohort cine CMR benchmark, CineMR achieves **35.9% pass@1** and **58.9% pass@4**, compared with **1.5% pass@1** for the Qwen3-VL-8B backbone and **0.0%** and **7.0% pass@1** for LLaVA-Med v1.5 and MedGemma-4B, respectively. Correct tool invocation reaches **99.8%** after GRPO, up from **78.9%** after SFT. Live tool outputs improve ventricular measurement accuracy by **20.4–23.7%** over direct model predictions, and removing all tools reduces pass@1 from **35.9%** to **27.9%**. These results highlight the importance of reliable tool use for quantitative cine CMR reasoning and support CineMR as a promising approach for assistive cardiac image assessment. Code, benchmark resources, and model weights are available at <https://github.com/AI-MIND-Lab/CineMR>.

1 Introduction

Cardiovascular magnetic resonance (CMR) is a reference standard for the noninvasive quantification of ventricular volumes, myocardial mass, and systolic function [Petersen et al., 2016, Schulz-Menger et al., 2020]. Clinical CMR interpretation integrates visual assessment with quantitative measurements of ventricular volumes, ejection fraction, myocardial mass, wall thickness, and regional wall motion, together with clinical history and complementary imaging findings [Arbelo et al., 2023, Schulz-Menger et al., 2020]. Quantitative criteria can support phenotype classification, but definitive diagnosis may also require clinical, electrocardiographic, genetic, coronary, and tissue-characterization information. The combination of multidimensional image analysis, quantitative measurement, and criterion-based interpretation makes CMR a compelling testbed for tool-augmented

LLaVA-Med v1.5 (Mistral-7B)	MedGemma (Google, 4B)	CineMR (Ours, Based on Qwen3-VL-8B)
Reasoning (text only): To analyze this cardiac cine MRI, I would first observe the left and right ventricular morphology and function. I will start by segmenting the cardiac structures and then measuring the volumes and wall thickness. Based on these measurements, I can formulate a hypothesis about the patient's cardiac condition and then verify it with additional imaging or clinical information. <think> Next, I would verify the hypothesis using quantitative measurements obtained from the MRI images. For example, I might use tools to segment the cardiac structures, ..., and analyze the trabeculation patterns. By comparing these measurements ... NO DEFINITIVE ASSESSMENT PRODUCED ✖ Question: Analyze this cardiac cine MRI and provide a comprehensive diagnostic assessment. Systematically evaluate the left and right ventricular morphology and function, identify any regional wall motion abnormalities, and arrive at the most likely diagnosis. Use the hypothesis verification approach: state your initial phenotypic hypothesis, then verify with quantitative measurements.	<think>The image shows a single axial slice of the heart. I need to assess the left and right ventricular morphology and function. I will start by segmenting the cardiac structures and then measuring the volumes and wall thickness. Based on these measurements, I can formulate a hypothesis about the patient's cardiac condition and then verify it with additional imaging or clinical information. <think> <tool_call>{"name": "segment_cardiac", "arguments": {"image": "image.png"}}</tool_call> <tool_result>{"left_ventricle": {"volume": 60, "wall_thickness": 12}, "right_ventricle": {"volume": 30, "wall_thickness": 8}, "atria": {"left_atrium": {"volume": 20}, "right_atrium": {"volume": 10}}}</tool_result> <think>The left ventricle volume is 60 ml, and the right ventricle volume is 30 ml. The left ventricular wall thickness is 12 mm, ..., and the right atrium volume is 10 ml. These measurements are within normal limits. I will now analyze the wall motion abnormalities. <think> UNSUPPORTED TOOL OUTPUT ✖ [... identical <think> block repeated 15 more times ...] <think>The patient's REPETITION GENERATION ✖	<think> Hypothesis: The imaging pattern suggests a dilated, globally hypokinetic phenotype. I need to quantify these observations with tools. <think> <think> I need to segment the cardiac structures to identify the left ventricle, right ventricle, and myocardium. <think> <tool_call> ["name": "segment_cardiac", "arguments": {}]</tool_call> <tool_result> {"status": "success", "segmentation": "...seg_mask"}</tool_result> <think> Segmentation complete. I can now compute quantitative measurements. <think> ... <think> Verification 1: LVEF = 29.8572637810542% → severely reduced (normal: 52-72%). ... Conclusion: DCM-compatible phenotype under the benchmark criteria. <think> ✓

Figure 1: Comparison of model outputs for the same cine CMR case, where the benchmark reference category is dilated cardiomyopathy (DCM). In this illustrative example, LLaVA-Med fails to provide a definitive assessment, whereas MedGemma produces unsupported tool-call content and repetitive text. In contrast, CineMR invokes executable tools to obtain quantitative measurements and evaluates them according to criteria encoded in the study-specific Rule-Based Cardiac Phenotype Classifier, thereby producing a traceable output grounded in explicit measurements and decision rules.

multimodal reasoning. This is particularly important because errors in upstream measurements can propagate to downstream classifications.

Recent medical VLMs have expanded multimodal question answering and generated-rationale capabilities, with reinforcement learning to optimize structured response generation [Lai et al., 2025, Pan et al., 2025, Su et al., 2025, Li et al., 2023, Sun et al., 2025]. However, general-purpose medical VLMs are not designed to perform the multidimensional image analysis required to derive quantitative CMR measurements reliably. As illustrated in Fig. 1, the evaluated baseline models may fail to produce a definitive assessment or generate repetitive text, unsupported tool outputs, or ungrounded quantitative values when prompted to interpret cine CMR. This limitation reflects the explicit multidimensional computations required for quantitative CMR assessment. For example, assessment of a DCM-compatible phenotype requires delineation of the left ventricle across multiple slices and cardiac phases, voxel-based calculation of ventricular volumes and ejection fraction, and interpretation using reference ranges and study criteria [Petersen et al., 2017]. Current VLMs cannot be assumed to reproduce this multistep quantitative pipeline reliably from image inputs alone.

Tool-augmented medical agents [Li et al., 2024, Fathi et al., 2025, Wang et al., 2026, Jiang et al., 2025] offer a promising direction, although many existing systems emphasize perceptual operations, such as segmentation visualization and region enlargement, rather than the explicit computation of clinically interpretable imaging measurements. Quantitative descriptors such as the septal-to-lateral wall-thickness ratio and mass-to-volume ratio may contribute to differentiating hypertrophic cardiomyopathy (HCM) from hypertensive heart disease (HHD), but they must be interpreted with additional morphological and clinical information; assessment of a DCM-compatible phenotype incorporates ventricular dilation and systolic dysfunction relative to appropriate sex-specific reference ranges [Petersen et al., 2017]. These measurements cannot be quantified reliably through visual inspection alone. These tasks require quantitative measurement in addition to visual interpretation, motivating tools that return explicit quantitative outputs. CineMR addresses this need by equipping a VLM with cardiac image-analysis tools and training it to invoke and integrate their outputs within an interleaved reasoning process.

Rule-Based Cardiac Phenotype Classifier-structured dataset. To make the supervision clinically traceable and systematically cover criterion-based reasoning, we organize the benchmark around a Rule-Based Cardiac Phenotype Classifier rather than relying only on direct image-to-label associations. The Rule-Based Cardiac Phenotype Classifier provides explicit diagnostic pathways linking quantitative measurements to reference answers and expected tool use. We derive the training data from a study-specific Rule-Based Cardiac Phenotype Classifier informed by published cardiomyopathy criteria, CMR reference studies, and explicitly documented project-specific thresholds [Arbelo et al., 2023, Petersen et al., 2017]. The Rule-Based Cardiac Phenotype Classifier covers ten harmonized outcome categories. Each question maps to a Rule-Based Cardiac Phenotype Classifier node annotated with its task-complexity level (L1–L6) and the tools expected to generate the reference answer. This structure encourages the model to follow criterion-based pathways and provides traceable supervision for both answer generation and tool use.

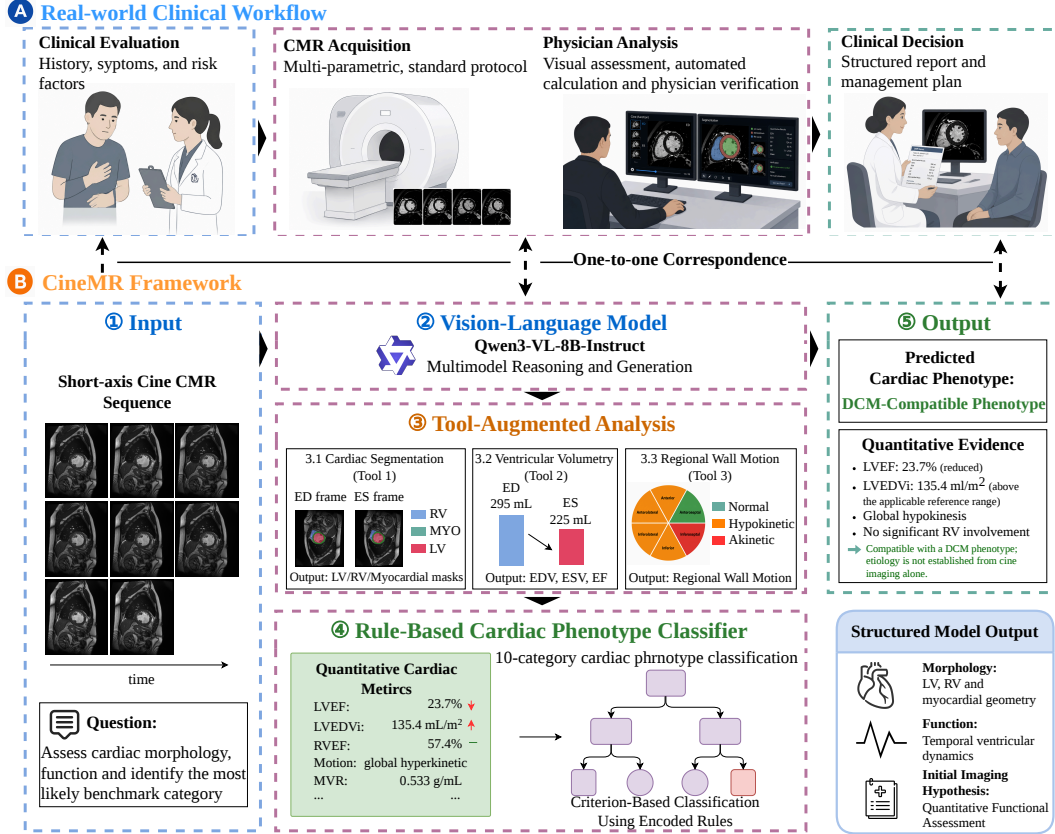


Figure 2: Overview of the CineMR framework, designed to mirror the real-world cardiologist workflow. Just as a clinician first inspects the cine images to form a visual impression, CineMR generates an initial phenotypic hypothesis via chain-of-thought reasoning. The clinician then orders quantitative analyses (segmentation, volumetry, wall thickness measurement); analogously, CineMR invokes a cardiology-specific quantitative tool suite to obtain precise metrics (EDV, ESV, EF, mass, MVR, wall thickness, sphericity index, regional wall motion). Finally, where the clinician interprets these measurements against ESC/AHA guideline thresholds to reach a diagnosis, CineMR routes them through a guideline-aligned Rule-Based Cardiac Phenotype Classifier with threshold-gated branching, producing an auditable reasoning chain that parallels the clinical diagnostic process from visual assessment to quantitative verification to evidence-based conclusion.

Cardiac image-analysis tool suite. We develop a cardiology-specific tool suite for quantitative cine CMR analysis. LV, RV, and myocardial segmentation is performed using a task-specific model fine-tuned from CineMA [Fu et al., 2025]. Voxel-based volumetry computes biventricular end-diastolic volume (EDV), end-systolic volume (ESV), and ejection fraction (EF). Morphometric analysis further derives myocardial mass, mass-to-volume ratio (MVR), maximal wall thickness, septal-to-lateral wall-thickness ratio, and sphericity index. Regional wall motion is assessed at the segment level using an AHA segment-based representation. These tools provide the VLM with segmentation-derived quantitative measurements. The measurements are used at the principal quantitative branches of the study-specific Rule-Based Cardiac Phenotype Classifier. This design allows the model to condition its responses on explicit quantitative evidence rather than on image features alone.

Two-stage training with conditional tool rewards. A conventional approach to tool-use training is supervised fine-tuning (SFT) on reference tool-interaction traces. SFT teaches the model the tool-call syntax, the interleaved reasoning format, and the association between different question types and their expected tools. However, SFT mainly imitates predefined trajectories and does not directly optimize whether the model selects tools appropriately during its own generation. We therefore further apply Group Relative Policy Optimization (GRPO) [Shao et al., 2024] to refine the model’s tool-use policy. The GRPO stage uses a conditional tool-use reward, assigning credit for tool invocation only

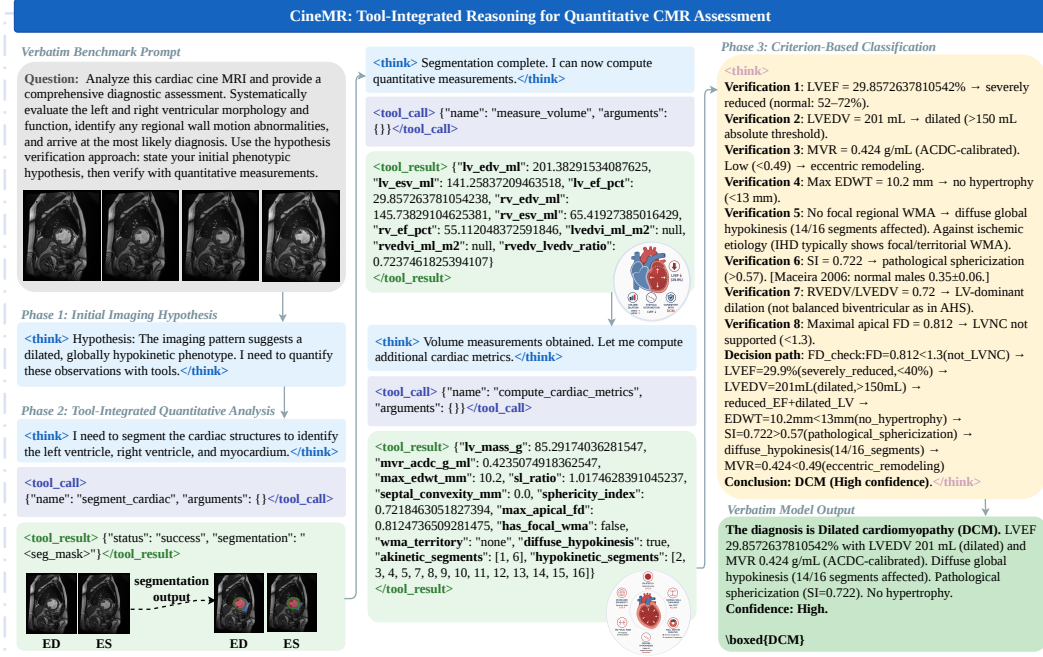


Figure 3: Representative verbatim CineMR rollout for a cine CMR case assigned to the DCM category. The figure shows the original benchmark prompt, model-generated tool calls, executable tool outputs, intermediate generated rationale, and final predicted category. CineMR invokes cardiac segmentation, ventricular volumetry, and morphometric analysis tools before applying the published and study-specific criteria encoded in the Rule-Based Cardiac Phenotype Classifier. Tool results are displayed at their native quantitative precision. The intermediate rationale and final response are reproduced verbatim and should not be interpreted as independently validated clinical conclusions.

when the final answer is correct. This encourages the model to learn when tool use is beneficial while reducing unnecessary or indiscriminate tool calls.

We evaluate CineMR on a multi-cohort benchmark constructed from ACDC [Bernard et al., 2018], M&Ms, and M&Ms-2 [Campello et al., 2021, Martín-Isla et al., 2023]. The benchmark spans ten harmonized outcome categories and six levels of task complexity (L1–L6). CineMR achieves 35.9% pass@1 and 58.9% pass@4 overall, compared with 1.5% pass@1 (4.5% pass@4) for the unfine-tuned Qwen3-VL-8B backbone, 7.0% pass@1 (18.0% pass@4) for MedGemma-4B-it, and 0.0% pass@1 (0.0% pass@4) for LLaVA-Med v1.5. On L3–L4 tasks requiring criterion-based quantitative assessment, CineMR achieves 58.7% pass@1 and 85.5% pass@4, whereas end-to-end multicategory diagnosis across all ten benchmark categories (L5) remains substantially harder, at 13.3% pass@1, indicating that CineMR is not yet a reliable standalone diagnostic system. GRPO also raises correct tool selection to 99.8% of gold tool-requiring items, up from 78.9% after supervised fine-tuning, and live tool outputs improve ventricular measurement accuracy by 20.4–23.7% over direct model predictions, highlighting the importance of reliable tool use for quantitative cine CMR reasoning and supporting tool-augmented VLMs as a promising approach for assistive cardiac image assessment.

2 Related Work

Tool-Augmented Medical Agents. Integrating multimodal large language models (MLLMs) with external tools can extend their capabilities beyond image and text generation. MMedAgent [Li et al., 2024], VILA-M3 [Nath et al., 2025], and AURA [Fathi et al., 2025] incorporate task-specific tools for operations such as segmentation, classification, and image annotation. MedAgent-Pro [Wang et al., 2026] and AgentMD [Jin et al., 2024] organize specialized models or tools within agent-based diagnostic or clinical risk-assessment workflows. Ophiuchus [Jiang et al., 2025] is particularly relevant because it uses a three-stage training strategy comprising supervised fine-tuning, self-reflection fine-tuning, and agentic Group Relative Policy Optimization [Shao et al., 2024] to train a

VLM to select and invoke image-analysis tools within an interleaved generation process. Relative to the requirements of quantitative CMR assessment, many of these approaches emphasize perceptual tools, such as segmentation visualization and region enlargement, rather than tools that return explicit imaging measurements. In addition, several systems rely primarily on prompting or supervised fine-tuning for tool selection. CineMR extends this paradigm by introducing quantitative tools that derive explicit quantitative measurements from segmentation outputs and by organizing training data around a study-specific Rule-Based Cardiac Phenotype Classifier designed to cover the encoded benchmark criteria systematically.

AI for Cardiac MRI Analysis. Deep learning methods have shown strong performance on several cine CMR analysis tasks. The ACDC challenge [Bernard et al., 2018], M&Ms challenge, and M&Ms-2 challenge [Campello et al., 2021, Martín-Isla et al., 2023] provide widely used benchmarks for cardiac segmentation. Representative segmentation approaches include nnU-Net [Isensee et al., 2021] and task-specific models fine-tuned from the CineMA foundation model [Fu et al., 2025]. Prior work has also developed automated pipelines for quality-controlled segmentation and quantitative characterization of cardiac function [Ruijsink et al., 2020], as well as radiomics models for characterizing cardiovascular risk-related imaging phenotypes [Cetin et al., 2020]. To our knowledge, prior work has not combined executable quantitative cine CMR analysis tools with an interleaved VLM generation process for criterion-based cardiac phenotype assessment. CineMR differs from these approaches by training a VLM to generate an initial imaging hypothesis, invoke executable tools for quantitative analysis, and integrate the returned measurements using criteria encoded in a study-specific Rule-Based Cardiac Phenotype Classifier.

3 Method

CineMR trains a vision-language model to perform criterion-based cardiac phenotype assessment by strategically invoking quantitative tools within an interleaved generation process. An overview of the proposed framework is shown in Fig. 2. We first formulate the diagnostic reasoning problem (§3.1), and then introduce the Rule-Based Cardiac Phenotype Classifier-structured dataset (§3.2), the quantitative tool suite (§3.3), and the two-stage training pipeline (§4).

3.1 Problem Formulation

Let $\mathcal{I}$ denote a cine CMR sequence, q a benchmark question, and $\mathcal{T} = \{t_1, \dots, t_K\}$ the available tool set. We model the task as an agentic generation process: at each turn i , the model produces reasoning text r_i or emits a tool invocation $\tau_i = (t_k, a_i)$; the runtime executes t_k on the imaging input I using arguments a_i and returns $\mathbf{o}_i = t_k(I, a_i)$ which is inserted into the generation context. The full response $y = (r_1, \tau_1, \mathbf{o}_1, r_2, \dots, r_n)$ ends with a predicted answer $\hat{d} \in \mathcal{D}$, where $\mathcal{D}$ may represent a numerical value, binary finding, or benchmark category, depending on the question type.

As shown in Fig. 3, this proceeds in three phases: (1) a `<think>` block forming an initial imaging hypothesis, (2) one or more `<tool_call>/<tool_result>` turns obtaining quantitative measurements, and (3) a narrative synthesis that evaluates the returned measurements using the published and study-specific criteria encoded in the Rule-Based Cardiac Phenotype Classifier before producing the final predicted category $\hat{d}$.

3.2 Rule-Based Cardiac Phenotype Classifier-Structured Dataset

Published cardiomyopathy guidelines and CMR reference studies contain quantitative criteria that can inform phenotype assessment. We encode a subset of these criteria, together with explicitly documented study-specific thresholds, into benchmark decision rules, including criteria related to LVEF, wall thickness, ventricular volumes, and regional wall motion. Several CMR findings relevant to cardiac phenotype assessment can be represented using quantitative criteria. However, definitive clinical diagnosis also incorporates history, symptoms, electrocardiography, genetics, loading conditions, coronary assessment, and tissue-characterization imaging. This design provides three advantages: (i) criterion-traceable supervision based on deterministic answer-generation rules rather than unconstrained LLM-generated labels, (ii) node-level annotations specifying the tools expected by the reference policy, and (iii) six task-complexity levels (L1–L6).

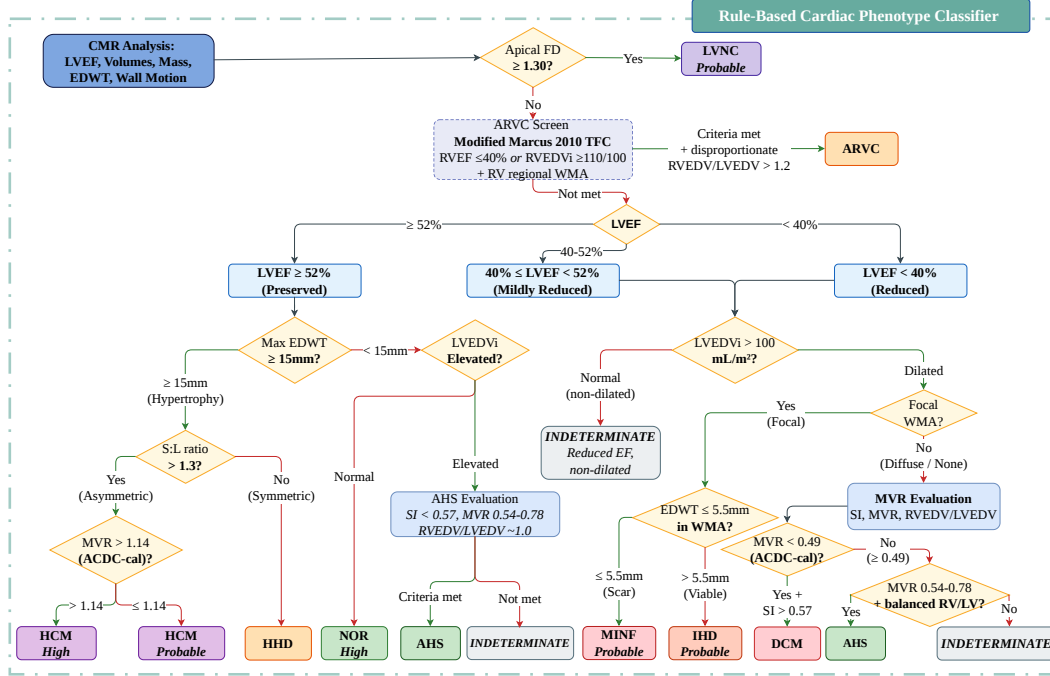


Figure 4: The tools produce the quantitative measurements required by the study-specific Rule-Based Cardiac Phenotype Classifier. Final classification includes the following categories: hypertrophic cardiomyopathy (HCM), athlete’s heart or physiological athletic remodeling, arrhythmogenic right ventricular cardiomyopathy (ARVC), left ventricular non-compaction (LVNC), ischemic heart disease (IHD), myocardial infarction (MINF), dilated cardiomyopathy (DCM), hypertensive heart disease (HHD), normal (NOR), and indeterminate.

In clinical practice, CMR interpretation integrates visual assessment with quantitative measurements and other available clinical and imaging information rather than exhaustively traversing a fixed Rule-Based Cardiac Phenotype Classifier [Schulz-Menger et al., 2020]. The decision rules represent the verification pathways encoded for this benchmark, while the model learns to select tools and integrate their outputs using the corresponding published and study-specific criteria. For each patient, routing the extracted metrics through the Rule-Based Cardiac Phenotype Classifier yields a candidate diagnosis, which is reconciled against the cohort’s independently confirmed clinical label: whenever the two disagree, the confirmed cohort label takes precedence and becomes the boxed reference answer. This reconciliation step overrode the classifier’s candidate diagnosis in only 2 of 3,320 patients, indicating that the encoded rules agree with the cohort’s confirmed diagnoses in the overwhelming majority of cases.

The Rule-Based Cardiac Phenotype Classifier in Fig. 4 assigns cases to ten harmonized benchmark categories: NOR, DCM, HCM, HHD, MINF, IHD, ARVC, athlete’s heart, LVNC, and INDETERMINATE. Each branching point uses either a published quantitative criterion or an explicitly documented project-specific threshold or calibration. LVEF classification uses sex-specific reference ranges reported by [Petersen et al., 2017]. The HCM-related benchmark rules are informed by the 2023 ESC cardiomyopathy guideline [Arbelo et al., 2023]. In adults, otherwise unexplained maximal LV wall thickness of ≥ 15 mm supports HCM, whereas a threshold of ≥ 13 mm may support HCM in first-degree relatives of patients with unequivocal HCM or in the presence of other compelling clinical or genetic evidence. The ARVC-related benchmark rules use the structural and functional CMR components of the 2010 Revised Task Force Criteria [Marcus et al., 2010]. For ACDC, MVR thresholds are adjusted using a project-specific factor of 0.913 to account for differences in the treatment of papillary muscles. This adjustment is informed by studies demonstrating the effects of papillary-muscle inclusion on LV mass and volume measurements [Vogel-Claussen et al., 2006, Gommans et al., 2016]. No corresponding adjustment is applied to M&Ms or M&Ms-2.

Task-complexity levels. We organize questions into six cognitive layers of increasing reasoning complexity:

Table 1: **Cardiology-specific tool suite.** Each tool maps to Rule-Based Cardiac Phenotype Classifier nodes requiring quantitative thresholds inaccessible to visual inspection.

Tool	Input $\rightarrow$ Output	Unlocked Metrics	Role
segment_cardiac	$\mathcal{I} \rightarrow s$	LV/RV/MYO masks	prerequisite
measure_volume	$s \rightarrow v$	EDV, ESV, EF, EDVi	computational
compute_cardiac_metrics	$(s, v) \rightarrow m$	mass, MVR, EDWT, S:L, SI	computational
wall_motion_analysis	$s \rightarrow w$	segment-level WMA	computational
select_cardiac_phases	$s \rightarrow (t_{ED}, t_{ES})$	temporal grounding	auxiliary
zoom_anatomy	$(t, target) \rightarrow ROI$	local visual detail	auxiliary

- **L1, numeric extraction:** direct retrieval of quantitative metrics, e.g., “What is the LVEF?” Tool use is required.
- **L2, binary finding assessment:** evaluation of a visual or anatomical finding, e.g., “Is the LV dilated?” Tool use is optional under the benchmark reference policy, except for templates that explicitly require quantitative measurements.
- **L3, criterion evaluation:** comparison with a published or study-specific threshold, e.g., “Does EDWT satisfy the HCM criterion of ≥ 15 mm?” Tool use is required.
- **L4, pairwise criterion-based classification:** distinction between two benchmark categories, e.g., HCM versus HHD. Multiple tools are required under the reference policy.
- **L5, multicategory assessment:** end-to-end prediction across all ten benchmark categories using a hypothesis-and-verification format.
- **L6, targeted differential assessment:** distinction between clinically confusable benchmark categories, e.g., DCM versus MINF.

The resulting 19 templates comprise 6, 4, 4, 2, 1, and 2 questions for L1, L2, L3, L4, L5, and L6, respectively. (Appendix B lists them all). These levels progress from quantitative measurement and criterion evaluation to multicategory and targeted differential assessment, with L3–L6 requiring increasingly selective and compositional tool use under the benchmark reference policy.

Metric provenance. We audited all 66 classifier thresholds for clinical traceability: 43 were grounded in guidelines or peer-reviewed studies, while 23 were project-specific thresholds with documented derivations and published precedents.

3.3 Cardiology-specific Quantitative Tool Suite

Clinical cardiac MRI diagnosis is fundamentally a *measurement-driven* process: ESC/AHA guidelines define diagnostic criteria through quantitative thresholds, such as $LVEF < 40\%$ for reduced systolic function and $EDWT \geq 15$ mm for HCM [Arbelo et al., 2023, Ommen et al., 2020]. Obtaining these measurements from raw imaging data requires multi-step computation, including anatomical delineation, voxel-wise integration, and derived ratio calculation, which current VLMs cannot reliably perform internally. Many prior tool-augmented medical agents [Li et al., 2024, Jiang et al., 2025] emphasize perceptual tools, such as segmentation visualization and region enlargement, rather than tools that return explicit quantitative imaging measurements. We therefore design a *quantitative* tool suite with $K = 6$ tools, organized as a dependency chain in which each tool’s output feeds subsequent computations. These tools produce the quantitative measurements required by the study-specific Rule-Based Cardiac Phenotype Classifier.

Segmentation tool. $segment_cardiac(\mathcal{I}) \rightarrow s$ uses a task-specific segmentation model fine-tuned from CineMA [Fu et al., 2025], a multiview convolutional-transformer masked-autoencoder foundation model pretrained on approximately 15 million cine CMR images from 74,916 UK Biobank studies, to produce frame-wise masks for the left ventricle (LV), right ventricle (RV), and myocardium (MYO). This tool is a prerequisite for downstream quantitative analyses, whose measurements are derived using standardized quantitative post-processing rather than unstructured visual estimation.

Volumetric tools. $measure_volume(s) \rightarrow v$ computes biventricular end-diastolic and end-systolic volumes, ejection fractions, and, when body surface area is available, indexed volumes using voxel-based volumetry across the short-axis stack. When body surface area is unavailable, indexed volumes are not reported. LVEF and indexed ventricular volumes are major quantitative features used by the study-specific Rule-Based Cardiac Phenotype Classifier [Arbelo et al., 2023].

LVEF determines the initial branch preserved, mildly reduced, or reduced while indexed volumes inform subsequent category-specific evaluations. Because reliable application of these criteria requires quantitative measurement, visual estimation alone is insufficient [Schulz-Menger et al., 2020].

`select_cardiac_phases(s)` $\rightarrow (t_{ED}, t_{ES})$ identifies end-diastolic and end-systolic frames from the LV cavity volume curve. Accurate phase identification is critical because incorrect ED/ES selection introduces systematic volume errors that propagate to all downstream metrics [Schulz-Menger et al., 2020]. This tool automates the clinically recommended approach of selecting ED as maximum and ES as minimum LV cavity volume.

Morphometric tool. `compute_cardiac_metrics(s, v)` $\rightarrow \mathbf{m}$ derives advanced diagnostic features, each linked to a specific branch of the diagnostic Rule-Based Cardiac Phenotype Classifier:

- *LV myocardial mass* ($V_{MYO} \times 1.05 \text{ g/mL}$): a standard CMR metric for assessing ventricular hypertrophy [Arbelo et al., 2023].
- *Mass-to-volume ratio (MVR)*: captures disproportionate myocardial thickening relative to chamber volume and supports discrimination between HCM and athlete’s heart [Czimbalmos et al., 2019].
- *Maximal end-diastolic wall thickness (EDWT)*: using an AHA segment-based representation and used to evaluate the encoded wall-thickness criteria.
- *Septal-to-lateral wall-thickness ratio (S:L)*: provides a quantitative descriptor of asymmetric septal hypertrophy and is used as one feature in the study-specific HCM-versus-HHD rule [Henry et al., 1973].
- *Sphericity index (SI)*: LV geometry and is used in the study-specific Rule-Based Cardiac Phenotype Classifier as a feature associated with spherical ventricular remodeling [Nakamori et al., 2017].

Wall motion tool. `wall_motion_analysis(s)` $\rightarrow \mathbf{w}$ evaluates regional wall motion by comparing endocardial radial contraction and myocardial thickening between ED and ES frames. Each AHA segment is classified as normal, hypokinetic, or akinetic. This tool identifies focal versus diffuse wall-motion patterns. A focal territorial abnormality may support an ischemic phenotype, whereas diffuse hypokinesis may be more compatible with a nonischemic dilated phenotype; however, cine wall motion alone does not establish etiology or myocardial scar.

4 Training Pipeline

We adopt a two-stage training paradigm inspired by recent agentic training approaches [Jiang et al., 2025, Shao et al., 2024], with modifications tailored to quantitative (rather than visual) tool use.

Stage 1: Cold-start SFT. We fine-tune Qwen3-VL-8B-Instruct [Bai et al., 2025] with LoRA (rank 16, $\alpha = 32$) on the training partition of the VQA benchmark using reference tool-use trajectories, whose displayed tool outputs are ground-truth-derived by default and may optionally be substituted with model-predicted measurements for trace realism; the boxed reference answer is always ground-truth-derived regardless. This stage trains the model to generate the tool-invocation syntax, follow the interleaved trace format, and associate question types with tools specified in the reference trajectories. We then augment this trajectory pool by rejection sampling: rollouts from the resulting policy are retained as additional SFT trajectories only if they yield a correct boxed answer, satisfy a tool-protocol adherence threshold, and, for classification items, are self-consistent across repeated samples.

Stage 2: Agentic GRPO. We apply Group Relative Policy Optimization (GRPO) [Shao et al., 2024], which samples a group of G responses $\{o_i\}_{i=1}^G$ from the current policy $\pi_{\theta_{\text{old}}}$ for each prompt q and optimizes:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{q, \{o_i\}} \left[\frac{1}{G} \sum_{i=1}^G \left(\min(r_i(\theta) \hat{A}_i, \text{clip}(r_i(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_i) - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right) \right] \quad (1)$$

where $r_i(\theta) = \pi_{\theta}(o_i \mid q) / \pi_{\theta_{\text{old}}}(o_i \mid q)$ is the importance ratio, $\hat{A}_i = (S_i - \mu_G) / \sigma_G$ is the group-normalized advantage computed from per-sample rewards S_i , and $D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$ is a KL-divergence penalty (estimated with the unbiased low-variance estimator of Shao et al. [2024]) that regularizes the policy toward a fixed reference policy π_{ref} , set to the Stage-1 SFT checkpoint and held frozen throughout Stage 2, with coefficient β . The composite reward is:

$$S = (1 - w_{\text{fmt}}) S_{\text{ans}} + w_{\text{fmt}} S_{\text{format}} + w_{\text{tool}} S_{\text{tool}} + w_{\text{live}} S_{\text{live}} - w_{\text{rep}} P_{\text{rep}} \quad (2)$$

where $S_{\text{ans}} \in [0, 1]$ scores answer accuracy (tolerance-scaled for numeric tasks, exact match for classification), $S_{\text{format}} \in [-0.5, 0]$ penalizes malformed outputs, $S_{\text{tool}} \in [-0.1, +0.3]$ is *conditional* on answer correctness (tool bonuses are awarded only when the final answer is correct, which is intended to discourage indiscriminate tool invocation without corresponding answer accuracy), S_{live} rewards successful live tool execution, and P_{rep} penalizes degenerate repetition. Default weights are $w_{\text{fmt}}=0.15$, $w_{\text{tool}}=0.12$, $w_{\text{live}}=0.03$, $w_{\text{rep}}=0.5$; these are hand-set defaults rather than the product of a systematic hyperparameter sweep, and we did not evaluate sensitivity to them.

5 Experiments

5.1 Setup

Datasets. We construct a multi-cohort cine CMR visual question answering benchmark from ACDC [Bernard et al., 2018], M&Ms [Campello et al., 2021], and M&Ms-2 [Martín-Isla et al., 2023]. The benchmark contains 11,531 question-answer pairs, including 7,020 training, 1,191 validation, and 3,320 test items across ten harmonized outcome categories. Each sample consists of a CMR volume, a benchmark question, a deterministic reference answer derived from the predefined decision rules, and annotations specifying the expected tool-use policy.

For each source cohort, we construct patient-level partitions based on the official dataset splits before generating questions and paraphrases, and subsequently merge the partitions across cohorts. Thus, all images, questions, paraphrases, tool annotations, and derived measurements associated with a given patient are restricted to a single partition, preventing patient-level data leakage. The validation split is used for checkpoint selection and epoch-level monitoring during both training stages. The test split is used only for final evaluation and is never used for model selection or hyperparameter tuning.

Metrics. A sampled response receives a score of 1 if its final boxed answer satisfies the item’s reference-answer specification and 0 otherwise. Numerical answers are evaluated using prespecified item-level tolerances, while categorical and binary answers are evaluated using normalized exact matching. All reported rollouts execute predicted tool calls live through the tool runtime.

We report pass@1 and pass@4 using the standard averaged pass@ k estimator of Chen et al. [2021], $\text{pass}@k := \mathbb{E}_{\text{items}} \left[1 - \binom{n-c}{k} / \binom{n}{k} \right]$, where c is the number of correct responses among n sampled rollouts for an item. With $n=4$, pass@1 is the mean fraction of correct rollouts per item, while pass@4 indicates whether at least one of the four rollouts is correct, averaged across items.

We further report two tool-use diagnostics (Figure 5a). The *correct tool call rate* is the fraction of gold tool-requiring items for which the predicted tool set exactly matches the gold tool set. The *correct tool return rate* is the fraction of executed gold-matched tool calls whose returned outputs agree with the ground truth within the prescribed tolerance.

Implementation. We fine-tune Qwen3-VL-8B-Instruct on tool-integrated reasoning traces using LoRA with rank $r=16$, scaling factor $\alpha=32$, and dropout 0.05. SFT uses AdamW in bf16 with a 4,096-token context window. Starting from the resulting checkpoint, we perform tool-integrated GRPO using LoRA with $r=64$ and $\alpha=128$, $G=4$ rollouts per prompt, a maximum response length of 1,024 tokens, and three training epochs. We use asymmetric clipping with $\epsilon_{\text{lo}}=0.2$ and $\epsilon_{\text{hi}}=0.3$, and set $\beta=0.03$ in Eq. 1. The reward weights in Eq. 2 are $w_{\text{fmt}}=0.15$, $w_{\text{tool}}=0.12$, $w_{\text{live}}=0.03$, and $w_{\text{rep}}=0.5$. These values are fixed heuristically and are not tuned using validation or test performance. Training is performed on NVIDIA A100 and H100 GPUs.

We report results for both the SFT checkpoint and the GRPO checkpoint. We compare against the unfine-tuned Qwen3-VL-8B-Instruct backbone [Bai et al., 2025] using the same CineMR tool-aware system prompt, so that this comparison isolates the effect of training from prompting. We additionally compare against the off-the-shelf medical VLMs LLaVA-Med v1.5 [Li et al., 2023] and MedGemma-4B-it [Søllergren et al., 2026]. SFT and GRPO are evaluated using four rollouts per item with temperature 0.7, from which pass@1 and pass@4 are computed.

Table 2: Held-out test performance by cohort ($n = 3,320$; ACDC: 696, M&Ms: 1,846, M&Ms-2: 778). All methods use $k = 4$ rollouts with temperature 0.7. SFT and GRPO correspond to the matched pre- and post-RL models under the same live-tool protocol. GRPO consistently improves performance over SFT across all cohorts, yielding the best overall results.

Method	ACDC			M&Ms			M&Ms-2			Overall		
	P@1	P@4	Mean	P@1	P@4	Mean	P@1	P@4	Mean	P@1	P@4	Mean
LLaVA-Med v1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
MedGemma-4B	5.2	16.1	5.5	7.7	19.1	7.4	6.7	17.0	6.3	7.0	18.0	6.7
Qwen3-VL-8B	1.3	4.9	1.4	1.4	4.2	1.4	1.9	4.6	1.4	1.5	4.5	1.4
SFT	15.2	28.3	15.6	26.5	41.6	26.3	24.2	42.0	24.4	23.6	38.9	23.6
GRPO	33.9	58.0	34.7	36.2	58.0	35.4	36.9	62.0	37.3	35.9	58.9	35.7

Table 3: Layer-wise performance of SFT (pre-RL) and GRPO (post-RL) with live tool execution, compared with the off-the-shelf MedGemma-4B-it and Qwen3-VL-8B baselines. Results are reported by cognitive layer on the held-out test set ($n = 3,320$). GRPO improves over SFT across most cognitive layers, with the largest gains observed in Layers 2–3.

Slice	MedGemma-4B			Qwen3-VL-8B			CineMR (SFT)			CineMR (GRPO)		
	P@1	P@4	Mean	P@1	P@4	Mean	P@1	P@4	Mean	P@1	P@4	Mean
Layer 1	1.9	7.5	2.2	2.9	8.3	2.6	6.5	20.2	6.4	10.8	31.5	10.1
Layer 2	17.6	38.8	17.0	0.5	1.7	0.4	27.6	33.0	27.3	65.3	92.0	65.3
Layer 3	8.4	23.3	7.2	0.6	1.9	0.6	55.2	83.7	55.8	60.9	86.7	60.7
Layer 4	2.8	15.0	4.2	0.0	1.9	0.5	50.5	78.5	52.1	45.8	78.5	50.0
Layer 5	0.0	0.0	0.0	0.0	0.0	0.0	12.9	33.3	12.6	13.3	32.5	13.0
Layer 6	0.0	1.5	0.4	0.0	1.5	0.4	37.9	50.0	35.6	37.9	37.9	37.1
All layers	7.0	18.0	6.7	1.5	4.5	1.4	23.6	38.9	23.6	35.9	58.9	35.7

5.2 Results

Table 2 presents the main results. GRPO raises pass@1 from 23.6% (cold-start SFT) to 35.9% overall, with the largest single-cohort gain on ACDC (15.2%→33.9%) and consistent gains on M&Ms (26.5%→36.2%) and M&Ms-2 (24.2%→36.9%). Under the same $k=4$ -averaged pass@1 protocol used throughout, LLaVA-Med produces no valid diagnoses (0.0%), MedGemma reaches 7.0%, and the un-fine-tuned Qwen3-VL-8B backbone reaches 1.5%; CineMR’s pass@1 is roughly $24\times$ higher than the un-fine-tuned backbone it is trained from, and more than $5\times$ higher than the strongest off-the-shelf baseline.

5.3 Ablations

Table 3 breaks down the main results by cognitive layer. On the clinical-criteria layers (L3–L4), where a measured value must be evaluated against a predefined clinical criterion, GRPO reaches 58.7% pass@1, compared with 54.5% for SFT and at most 3.0% for either off-the-shelf baseline. This large gap suggests that explicit tool-integrated quantitative reasoning is particularly important when assessment depends on measured evidence rather than qualitative visual appearance alone. On the full-diagnosis layers (L5–L6), GRPO (18.6%) only marginally improves over SFT (18.3%), consistent with the compounding difficulty of multi-step reasoning, where errors in upstream segmentation or measurement can propagate through subsequent decision steps. Thus, CineMR’s gains are not uniform across visual question answering tasks, but are concentrated on tasks that require converting cine MRI into explicit quantitative evidence and evaluating that evidence against clinical criteria.

Figure 5a further shows that GRPO primarily improves the precision of tool selection. GRPO predicts the exact gold tool set on 99.8% of gold tool-requiring items, compared with 78.9% for SFT. Figure 5b breaks this down further: across the four volumetric and ejection-fraction question types, the live measure_volume tool’s own returned value is correct substantially more often (28–34%) than the model’s final boxed answer (4–11%), indicating that part of this gap reflects how the model integrates a returned measurement rather than the measurement’s own accuracy.

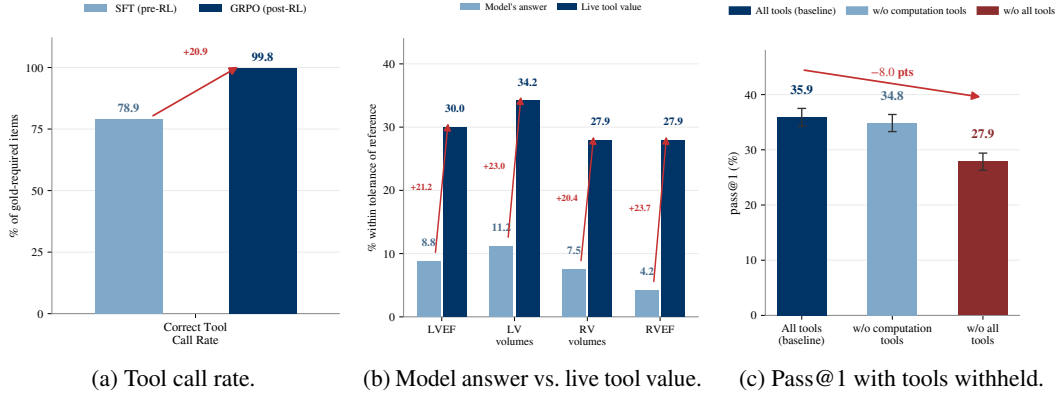


Figure 5: **Tool-use diagnostics and tool-withholding ablation.** (a) Correct tool-call rate for SFT and GRPO on gold tool-requiring items. (b) GRPO accuracy of direct model predictions versus live `measure_volume` outputs on ventricular volume and ejection-fraction questions. (c) GRPO pass@1 on the held-out test set ($n = 3,320$, $k = 4$) with full tool access, without computation tools, and without any tools; error bars show 95% bootstrap confidence intervals. GRPO raises correct tool invocation from 78.9% to 99.8%; live tool outputs improve measurement accuracy by 20.4–23.7%, and removing all tools lowers pass@1 from 35.9% to 27.9%.

We next test how much these quantitative outputs contribute to final performance. Withholding the three metric-producing tools at inference time, while retaining segmentation, wall-motion analysis, and visual zoom, reduces pass@1 from 35.9% to 34.8% (Figure 5c). Without explicit measurements, the model must instead rely on visual cues and learned priors for questions governed by numerical criteria. Removing all tools, including segmentation, further reduces pass@1 to 27.9%. The 95% bootstrap confidence interval for this setting does not overlap with that of the full-tool configuration, indicating that the degradation is unlikely to be attributable to sampling variation alone. Together, these results show that externally computed image-derived evidence contributes information beyond what the VLM recovers from visual reasoning alone.

6 Conclusion

We presented CineMR, a tool-augmented vision-language framework for quantitative cine CMR assessment. CineMR combines visual interpretation with dedicated tools for segmentation, phase selection, ventricular volumetry, morphometric measurement, and regional wall-motion analysis, and maps the resulting evidence to benchmark categories using an explicitly encoded Rule-Based Cardiac Phenotype Classifier. On the constructed multi-cohort benchmark, supervised fine-tuning and reinforcement learning substantially improved performance relative to the unfine-tuned backbone, while access to quantitative tools provided an additional, more modest improvement over the tool-ablated configuration. These findings demonstrate the potential of integrating vision-language models with explicit quantitative computation for traceable assessment of cine CMR examinations. However, the benchmark categories and study-specific rules do not constitute complete clinical diagnoses, and the framework has not been validated for autonomous clinical use. Future work should evaluate external generalizability, measurement robustness, calibration, and clinical utility across institutions, scanner vendors, patient populations, and CMR acquisition protocols.

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A Acronym Glossary

Table 4 collects every abbreviation used in the main text and appendix. Each is also spelled out at first use in the running text; this table is provided as a single point of reference.

Table 4: Acronym glossary.

Acronym	Definition
LV / RV	Left / right ventricle (ventricular)
LVEF / RVEF	Left / right ventricular ejection fraction
LVEDV / LVESV	LV end-diastolic / end-systolic volume
RVEDV / RVESV	RV end-diastolic / end-systolic volume
EDWT	End-diastolic wall thickness
S:L ratio	Septal-to-lateral wall thickness ratio
SI	Sphericity index
MVR	Mass-to-volume ratio (LV mass / LVEDV)
WMA	Wall motion abnormality
BSA	Body surface area
HCM	Hypertrophic cardiomyopathy
HHD	Hypertensive heart disease
DCM	Dilated cardiomyopathy
ARVC	Arrhythmogenic right ventricular cardiomyopathy
MINF	Myocardial infarction (ischemic scar)
IHD	Ischemic heart disease
AHS	Athlete’s heart syndrome
LVNC	Left ventricular noncompaction
NOR	Normal (no structural/functional abnormality)
rTFC	(2010) Revised Task Force Criteria for ARVC
ESC / AHA	European Society of Cardiology / American Heart Association
SCMR	Society for Cardiovascular Magnetic Resonance
CMR	Cardiac magnetic resonance

B Evaluation Question Templates

Section 3 (“Cognitive layer hierarchy”) defines the six layers and gives one illustrative example question per layer. Table 5 lists the complete set of **19 question templates** used to construct the ACDC / M&Ms / M&Ms-2 benchmark (Section 5.1), each mapped to its cognitive layer, tool requirement, and the tools it invokes when tools are used. Each template’s literal question text below is the *canonical* form; the released dataset additionally includes two LLM-generated paraphrase variants per base question (rephrasing, context enrichment, and specificity variation) for lexical diversity, so a given evaluation item may appear as a paraphrase of, rather than verbatim match to, the text shown here. The canonical templates, their answer-derivation functions, and applicability conditions are implemented in the released code, which is the authoritative source for exact question text, answer formats, and clinical citations.

Table 5: All 19 evaluation question templates, by cognitive layer. **Tools:** whether tool use is required (✓) or optional (–) for that template; a template may override its layer’s default (e.g. L2_is_asymmetric requires tools because the septal:lateral ratio must be computed).

Template ID	Layer	Tools	Question
L1_lvef	L1	✓	What is the left ventricular ejection fraction (LVEF) for this patient? Report as a percentage and indicate whether it is normal, mildly reduced, reduced, or severely reduced.
L1_lvedv	L1	✓	Report the left ventricular end-diastolic volume (LVEDV) and end-systolic volume (LVESV) for this cardiac MRI study, including BSA-indexed values if anthropometric data are available.
L1_rvedv	L1	✓	What are the right ventricular end-diastolic volume (RVEDV), end-systolic volume (RVESV), and RV ejection fraction (RVEF) for this patient?
L1_lv_mass	L1	✓	Calculate the left ventricular myocardial mass from this cardiac MRI segmentation. Report the absolute LV mass in grams.
L1_rvef	L1	✓	What is the right ventricular ejection fraction (RVEF) for this patient? Is it within normal limits?
L1_max_edwt	L1	✓	What is the maximum end-diastolic wall thickness (max EDWT) of the left ventricle in this cardiac MRI, and does it exceed the threshold for hypertrophy?
L2_is_dilated	L2	–	Is the left ventricle dilated in this cardiac MRI? Justify your answer using volumetric measurements.
L2_is_hypertrophied	L2	–	Does this patient demonstrate left ventricular hypertrophy? Report the maximum wall thickness and indicate whether it is within normal limits.
L2_is_asymmetric	L2	✓	Is the left ventricular hypertrophy asymmetric? Compute the septal-to-lateral wall thickness ratio.
L2_has_wma	L2	–	Does this cardiac MRI demonstrate regional wall motion abnormalities? If so, which AHA segments are affected and which coronary territory is involved?
L3_hcm_threshold	L3	✓	Does the maximum end-diastolic wall thickness meet the ESC 2023 criterion for hypertrophic cardiomyopathy (≥ 15 mm for sporadic cases)? Also assess whether the ≥ 13 mm familial HCM criterion is met.
L3_arvc_criteria	L3	✓	Do the RV volumetric measurements meet the 2010 Revised Task Force Criteria (rTFC) for ARVC? Assess both major and minor CMR criteria for RVEDVi and RVEF.
L3_minf_scar	L3	✓	In the segment(s) with regional wall motion abnormality, does the end-diastolic wall thickness meet the criterion for probable transmural scar (≤ 5.5 mm)? Assess the likelihood of chronic myocardial infarction versus viable hibernating myocardium.
L3_dcm_si	L3	✓	Is the left ventricular sphericity index elevated above the threshold for pathological sphericization (>0.57)? Compute the sphericity index from the short-axis stack.
L4_hcm_vs_hhd	L4	✓	This patient has left ventricular hypertrophy with max EDWT ≥ 13 mm. Is the hypertrophy pattern more consistent with hypertrophic cardiomyopathy (HCM) or hypertensive heart disease (HHD)? Use the S:L ratio, septal convexity, and mass-to-volume ratio in your assessment.
L4_dcm_vs_ahs	L4	✓	The left ventricle is dilated with reduced or borderline ejection fraction and no focal wall motion abnormality. Distinguish between dilated cardiomyopathy (DCM) and athlete’s heart syndrome (AHS) using the sphericity index, mass-to-volume ratio, and biventricular volume balance.
L5_full_diagnosis	L5	✓	Analyze this cardiac cine MRI and provide a comprehensive diagnostic assessment. Systematically evaluate the left and right ventricular morphology and function, identify any regional wall motion abnormalities, and arrive at the most likely diagnosis. Use the hypothesis-verification approach: state your initial phenotypic hypothesis, then verify with quantitative measurements.
L6_differential_dcm_minf	L6	✓	This patient has reduced left ventricular ejection fraction and ventricular dilation. The key differential is dilated cardiomyopathy (DCM) versus ischemic heart disease with prior myocardial infarction (MINF). What quantitative features allow you to distinguish between these two diagnoses?
L6_differential_hcm_athlete	L6	✓	This patient is a competitive athlete with increased left ventricular wall thickness (max EDWT in the 13–15 mm grey zone). Distinguish between hypertrophic cardiomyopathy (HCM) and physiological adaptation (athlete’s heart) using mass-to-volume ratio, septal shape, and cavity dimensions.

C Limitations and Broader Impact

Segmentation dependency. CineMR relies on a dedicated short-axis segmentation stage to derive quantitative cardiac metrics, including ventricular volumes, ejection fractions, myocardial mass, and wall thickness. Because downstream benchmark classification depends on these segmentation-derived measurements, segmentation errors can propagate to the final predicted category. In our error analysis, under- or over-segmentation of the left ventricular cavity was a frequent failure mode in L5 full-assessment tasks and could direct the Rule-Based Cardiac Phenotype Classifier toward an incorrect branch even when the subsequent reasoning process was otherwise consistent with the extracted measurements.

Toward segmentation-free quantitative reasoning. As multimodal foundation models and end-to-end learning methods advance, future systems may estimate quantitative cardiac measurements directly from imaging data within an integrated reasoning framework, potentially bypassing explicit segmentation. Such approaches may reduce segmentation-related error propagation; however, their measurement accuracy, robustness, and generalizability across acquisition protocols and scanner vendors will require dedicated validation.

Broader impact. By recording tool calls, quantitative outputs, and criterion-based classification paths, CineMR provides more inspectable evidence traces than an end-to-end prediction alone. With prospective external validation, the framework could potentially support quantitative cine CMR assessment in settings where specialist interpretation is limited. However, CineMR is not intended for autonomous clinical use. Its outputs should be reviewed by qualified clinicians, and overreliance on automated results particularly when upstream segmentation is inaccurate could contribute to assessment errors if the system were deployed without appropriate validation, quality control, and safeguards.

Additional limitations. This retrospective study uses public research datasets and primarily evaluates short-axis cine CMR. Several benchmark categories incorporate study-specific rules or thresholds and therefore should not be interpreted as complete clinical diagnoses. Future work should include prospective external validation across institutions, scanner vendors, and patient populations and assess calibration, subgroup performance, and robustness to variations in image quality and acquisition protocols.